

JEE MAIN REVISION MASTER: SOME BASIC CONCEPTS OF CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Mole Concept & Stoichiometry	12 PYQs	Stoichiometry
Moderate	Concentration Terms (Molarity, Molality, Mole Fraction)	4 PYQs	Concentration

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Mole Concept & Stoichiometry	$n = \frac{w}{M} = \frac{N}{N_A} = \frac{V_{\text{STP}}}{22.4}$, $\text{Combustion: } x\text{C}_x\text{H}_y + \left(x + \frac{y}{4}\right)\text{O}_2 \rightarrow x\text{CO}_2 + \frac{y}{2}\text{H}_2\text{O}$
Concentration Terms (Molarity, Molality, Mole Fraction)	$M = \frac{w_B \times 1000}{M_B \times V_{\text{mL}}}$, $m = \frac{w_B \times 1000}{M_B \times W_A}$, $x_B = \frac{n_B}{n_A + n_B}$, $\text{ppm} = \frac{w_B}{w_{\text{soln}}} \times 10^6$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mole Concept & Stoichiometry (High Yield — 12 PYQs)

Core Principle: Mole relationships, mass-volume stoichiometry, empirical vs molecular formulas, and limiting reagent identification.

■ **Caution / JEE Pitfall:** Always balance the chemical reaction before calculating limiting reagent moles/stoichiometric coefficient ratios.

■ **Speed Trick:** The reactant with the SMALLEST ratio of (Initial Moles / Stoichiometric Coefficient) is the limiting reagent.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Match List-I with List-II and select the correct option (Based on mole concept): List-I List-II (a) 2 moles of ethene (i) 11.2 L volume ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "From the given information, select the suitable law of chemical combination: Cupric Carbonate % of Cu % of C % of O Natural Sample 51.35 9.7..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "Select the INCORRECT statement/s from the following: (a) 22 books have infinite significant figures (b) In the answer of calculation \$2.5 \times \dots\$ [Ans: N/A]

3.2 Concentration Terms (Molarity, Molality, Mole Fraction) (Moderate Yield — 4 PYQs)

Core Principle: Formulas for solution concentrations, temperature dependence, and dilution relation $M_1 V_1 = M_2 V_2$.

■ **Caution / JEE Pitfall:** Molarity (M) and Normality (N) change with temperature (due to volume expansion), while Molality (m) and Mole Fraction (x) are temperature INDEPENDENT.

■ **Speed Trick:** Direct conversion formula: $m = (1000 M / 1000 d - M M_B)$, where d is solution density in g/mL.

Representative Source PYQs:

- **[2025 Shift] Q#3 (MCQ):** "Which of the following methods of expressing concentration are unitless?..." **[Ans: A]**
- **[2022 Shift] Q#2 (MCQ):** "An aqueous solution of alcohol contains 18g of water and 414g of ethyl alcohol. The mole fraction of water is..." **[Ans: N/A]**
- **[2020 Shift] Q#1 (MCQ):** "A gas mixture contains 25%He and 75%CH₄ by volume at a given temperature and pressure. The percentage by mass of methane in the mixture..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match List-I with List-II and select the correct option (Based on mole concept): List-I List-II (a) 2 moles of ethene (i) 11.2 L volume at STP (b) Mass is equal to 66 g (ii) 56 g (c) 1 g of H₂ (iii) 12.04×10^{23} Molecules (d) 2 moles of water vapours (iv) 1.5 mole of CO₂

- (A) a-ii, b-iv, c-i, d-iii
- (B) a-iii, b-i, c-iv, d-ii
- (C) a-i, b-iv, c-ii, d-iii
- (D) a-ii, b-iii, c-i, d-iv

Correct Answer: None

Solution Analysis: For mole concept, we use these basic relations: 1 mole of any substance contains 6.02×10^{23} particles. 1 mole of any gas at STP occupies 22.4 L. Also, Mass=Number of moles $\times$ Molar mass For (a), ethene is C₂H₄. Its molar mass is $2 \times 12 + 4 \times 1 = 28 \text{ g mol}^{-1}$ So, mass of 2 moles of ethene is $2 \times 28 = 56 \text{ g}$ Hence, (a) $\rightarrow$ (ii) For (b), mass is 66 g. This matches carbon dioxide, CO₂, because molar mass of CO₂ is $12 + 2 \times 16 = 44 \text{ g mol}^{-1}$ Number of moles of

Benchmark Problem 2: KCET 2025 (Q#3)

Which of the following methods of expressing concentration are unitless?

- (A) Mole fraction and Mass percent (W/W) **[Correct]**
- (B) Molality and Mole fraction
- (C) Mass percent (W/W) and Molality
- (D) Molality and Molarity

Correct Answer: A

Solution Analysis: Option A is correct: mole fraction and mass percent (w/w) are both unitless. Mole fraction $x_i = \frac{n_i}{\sum n_j}$ It's simply a ratio of moles, so no units. Mass percent (w/w) $\text{mass \%} = (\text{mass of solute} / \text{total mass of solution}) \times 100\%$ Again, a ratio of masses, giving a percentage without units. By contrast, molality has units of mol/kg and molarity has mol/L, so neither is unitless.